902 43500 HOMEWORK 7

All problems are in the book: Michael Sipser, *Introduction to the Theory of Computation*, 2nd Edition, PAWS Publishing Company, 2005.

Solution by Cheng-Chung Li

Problem 1 Show that NP is closed under union and concatenation
Ans. The first part is to prove NP is closed under union. For any two NP-languages L_1 and L_2, let M_1 and M_2 be the NTMs that decide them in polynomial time. We construct a NTM M that decides the union of L_1 and L_2 in polynomial time:
M="On input $\langle w \rangle$:

1. Run M_1 on w. If it accepts, *accept*.

2. Run M_2 on w. If it accepts, *accept*. Otherwise, *reject*."

M is an NTM since M_1 and M_2 are both NTMs. Also, M accepts w if and only if either M_1 or M_2 accepts w. Therefore, M decides the union of L_1 and L_2. Since both stages take polynomial time, the algorithm runs in polynomial time.

The second part is to prove NP is closed under concatenation. According to previous assumption, let us construct an NTM N that decides the concatenation of L_1 and L_2 in polynomial time:
N="On input $\langle w \rangle$:

1. For each way to cut w into substrings $w = w_1w_2$:

2. Run M_1 on w_1.

3. Run M_2 on w_2. If both accept, *accept*; otherwise continue with the next choice of w_1 and w_2.

4. If w is not accepted after trying all possible cuts, *reject*."

N is an NTM since M_1 and M_2 are both NTMs. Also, N accepts w if and only if w can be expressed as w_1w_2 such that M_1 accepts w_1 and M_2 accepts w_2. Therefore, N decides the concatenation of L_1 and L_2. Since stage 2 runs in polynomial time and it is repeated for at most $O(n)$ time, the algorithm runs in polynomial time.

$\square$

Problem 2 Show that NP is closed under the star operation.
Ans. Please see the page 301 in the textbook. $\square$
Problem 3 Let $DOUBLE{-}SAT = \{\langle \phi \rangle | \phi \text{ has at least two satisfying assignments}\}$. Show that $DOUBLE - SAT$ is NP-complete.
Ans. First, we have to prove $DOUBLE - SAT$ is in NP. On input ϕ, a nondeterministic polynomial time machine can guess two assignments and accept if both assignments satisfy ϕ. Thus this problem is in NP.

Second, we show that $SAT \leq_p DOUBLE-SAT$. The following TM M computes the polynomial time reduction f.
M="On input $\langle \phi \rangle$, a Boolean formula with variables $x_1, x_2, \ldots, x_m$.

1. Let ϕ' be $\phi \wedge (x \vee \overline{x})$, where x is a new variable.

2. Output $\langle \phi' \rangle$."

If $\phi \in SAT$, ϕ' has at least two satisfying assignments because we can obtain two assignments from the original assignment of ϕ by changing the value of x. If $\phi' \in DOUBLE-SAT$, ϕ is also satisfiable, because x does not appear in ϕ. Therefore, $\phi \in SAT$ if and only if $f(\phi) \in DOUBLE-SAT$.

$\square$